

Optical path difference measurements using symmetric dual-comb interferometry

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Abstract We present a common-path dual-comb technique for long-distance fiber ranging, addressing coherence and ambiguity limitations. Spectral phase retrieval combined with multi-Nyquist-zone processing enables measurements up to 102 km with 5.6 mm resolution. ©2026 The Author(s)

Introduction

Accurate optical path measurement over long distances is a fundamental requirement in fields such as fiber sensing, precision metrology, and structural health monitoring [1]. Among the different approaches, dual-comb interferometry has established itself as a highly effective technique, as it enables direct mapping of optical frequencies into the radio-frequency domain while preserving phase information, thus allowing precise and high-resolution distance measurements with low-bandwidth detection and simple setup configurations [2].

Conventional dual-comb ranging implementations are typically based on an asymmetric configuration, where two frequency combs propagate along different optical paths before interference. In this framework, the achievable non-ambiguity range (NAR) is fundamentally limited by the comb repetition rate f_R [3]. Despite its capabilities, this approach entails inherent compromises between measurement range, resolution, and acquisition speed. In addition, when extended to long distances, the independent propagation of the combs makes the system highly susceptible to environmental perturbations, leading to differential phase noise that accumulates with distance and ultimately limits the achievable accuracy [3]. In this context, maintaining a high degree of mutual coherence between the combs becomes critical, typically requiring complex phase-locking schemes or ultra-stable laser sources [4].

In this work, we propose a common-path dual-comb ranging scheme for long-distance high-resolution fiber measurements. This configuration intrinsically suppresses relative phase noise and significantly relaxes the coherence requirements of the system, enabling the use of less demanding laser sources while preserving phase-sensitive detection capabilities. Building on a previously demonstrated architecture [5], we combine spectral phase retrieval with a multi-Nyquist-

zone processing strategy to extract optical path differences (OPDs). In dual-comb systems, two characteristic frequency scales coexist: the repetition rate f_R , which defines the spacing of the comb lines, and the offset δf which governs the temporal mapping between optical and radio-frequency domains. These two scales inherently define different measurement regimes in terms of range and resolution. In this approach, the non-ambiguity range is determined by the inverse of the comb spacing offset δf , which entails an improvement equal to the dual-comb compression factor ($CF=f_R/\delta f$) with respect to conventional, asymmetric configurations, while maintaining the resolution proportional to the inverse of f_R . The proposed method allows simultaneous access to coarse and fine estimations of the optical path length, leading to sub-centimeter resolution measurements over fiber spans up to 100 km.

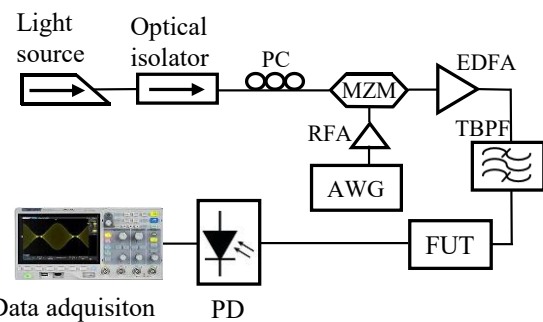


Fig. 1: Experimental setup. PC: Polarization controller; EDFA: Erbium-doped fiber amplifier; TBPF: Tunable band-pass filter RFA: Radio-frequency amplifier; FUT: Fiber under test; PD: Photodetector. The rest of acronyms are defined in the body of the manuscript.

Methods and experimental setup

The experimental implementation relies on the common-path dual-comb architecture previously reported in [5], here adapted for long-distance ranging, as shown in Fig. 1.

The optical frequency combs originate from a

single continuous-wave laser source whose central frequency is dithered up to have an equivalent linewidth of approximately 500 MHz, representative of a low-coherence laser. The optical carrier is modulated by a Mach-Zehnder modulator (MZM), driven by an electrically synthesized waveform generated using an arbitrary waveform generator (AWG). This waveform encodes the superposition of two frequency combs in the time domain (with repetition rates f_R and $f_R + \delta f$, respectively), enabling dual-comb generation within a single electro-optic modulation stage. The optical spectrum spans a bandwidth of $BW = 5 \text{ GHz}$, the repetition rate $f_R = 20 \text{ MHz}$ and the frequency offset is set to $\delta f = 1 \text{ kHz}$. After transmission of the two combs through the target fiber, the signal is detected by a single photodetector and digitized, yielding a multi-heterodyne interferogram in the radio-frequency domain. The acquisition time is set to 50 ms for all measurements.

Owing to the common-path propagation, phase fluctuations accumulated along the fiber link are shared by both combs and therefore suppressed in the beating process. This intrinsic noise cancellation enables stable phase-sensitive measurements even when using low-coherence optical sources, eliminating the need for complex phase-locking schemes.

The proposed ranging method is based on the extraction of spectral phase information from the multi-heterodyne signal. The digitized interferogram is mapped into the frequency domain via a fast Fourier transform (FFT), where the radio-frequency comb lines are resolved. The spectrum is subsequently segmented into Nyquist zones (NZs) by digital filtering, and only the phase of each comb line is retained for further processing. Notably, all Nyquist zones are extracted from the same acquired interferogram, avoiding the need for multiple measurements or system reconfiguration.

Distance estimation relies on the linear dependence between the OPD and the spectral phase evolution across frequency. When comparing two measurements corresponding to different fiber lengths, the resulting differential phase exhibits a linear trend whose slope is directly proportional to the OPD, as illustrated in Fig 2. In the first Nyquist zone, this phase evolution is sampled over a frequency grid defined by multiples of the comb offset $m \cdot \delta f$, where m is the line index. Under these conditions, the OPD can be retrieved from the phase slope as

$$\Delta L = \frac{b \cdot c_0}{2\pi \cdot n \cdot \delta f}, \quad (1)$$

being b the phase slope, c_0 the speed of light in a vacuum, and n the refractive index of the fiber. The measurable range in this zone is constrained by the phase ambiguity interval $[-\pi, \pi]$, leading to a maximum unambiguous distance inversely

proportional to δf , while the minimum detectable variation is determined by the phase estimation sensitivity. As a result, the first NZ provides a coarse but unambiguous estimate of the OPD over an extended range.

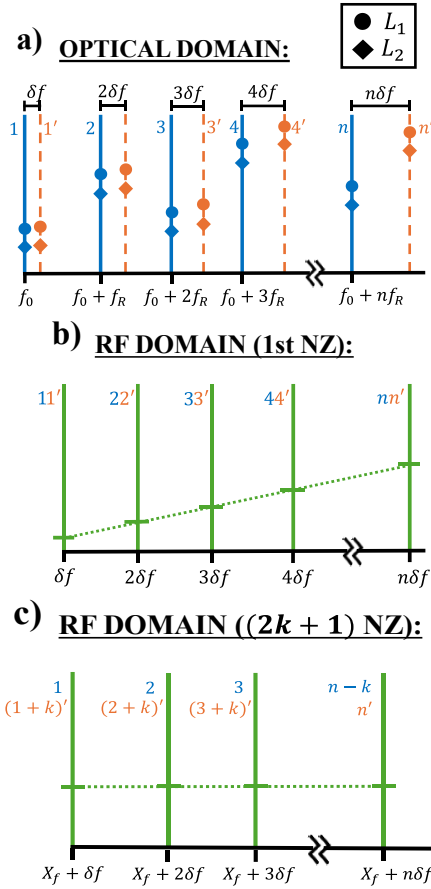


Fig. 2: **a)** Optical common-path combs after reaching the end of the FUT, where the phase at every line for 2 different optical paths (L_1 and L_2) are highlighted with different shapes; **b)** After photodetection; at 1st NZ, the slope of the phase gives all information for the OPD calculation. The colored numbers indicate the optical comb lines that generate each RF tone. **c)** After photodetection, where $X_f = k f_R + k \delta f$. The phase of higher-order NZs minus the phase of the 1st NZ provides a constant phase value already proportional to the OPD, as detailed in the body of the manuscript.

Higher-order Nyquist zones exhibit a different behaviour. Their frequency grids are shifted by integer multiples of the repetition rate, following $k \cdot f_R + m \cdot \delta f$, where k is linked to the order of the NZ as $k = \frac{NZ-1}{2}$ in the odd Nyquist zones considered in this work ($k = \frac{NZ}{2}$ for even NZ). By subtracting the phase evolution obtained in the first NZ, the contribution associated with $m \cdot \delta f$ is removed, effectively transforming the frequency grid into an equidistant set defined by $k \cdot f_R$. The phase profile then becomes constant across frequency, and its value can be readily obtained as its average quantity. The OPD is then obtained from an equivalent expression to Eq. (1) where $k \cdot f_R$ replaces δf , and the mean horizontal value

mentioned replaces the slope b . Thus, the resolution in higher-order NZ is improved in a factor $k \cdot f_R / \delta f$ with respect to that in the first one, at the expense of a reduced non-ambiguity range, now limited by $k \cdot f_R$. However, the coarse estimation provided by the first NZ enables unambiguous determination of the number of 2π phase cycles in higher-order zones.

By combining both contributions, the OPD can be reconstructed over extended distances with high precision. In this work, the processing is extended up to the 3rd NZ, limited by the detection bandwidth, enabling millimeter resolution while preserving a large measurement range. The minimum detectable OPD variation is ultimately limited by the phase estimation accuracy. In practice, additive noise does not constitute the dominant limitation, as stable measurements are obtained after averaging a small number of interferograms. Instead, the resolution is determined by the minimum resolvable phase variation, which is set by the uncertainty of the phase across the comb lines. This uncertainty is governed by the finite spectral resolution imposed by the acquisition time, the spacing of the frequency grid δf or f_R , and the number of comb lines contributing to the estimation.

Results

Fig. 3 shows the retrieved interferograms and the measured phase in the 1st and 3rd NZs (the latter corrected using the phase of the 1st one) in a representative case of ~ 4 km OPD. Tab. 1

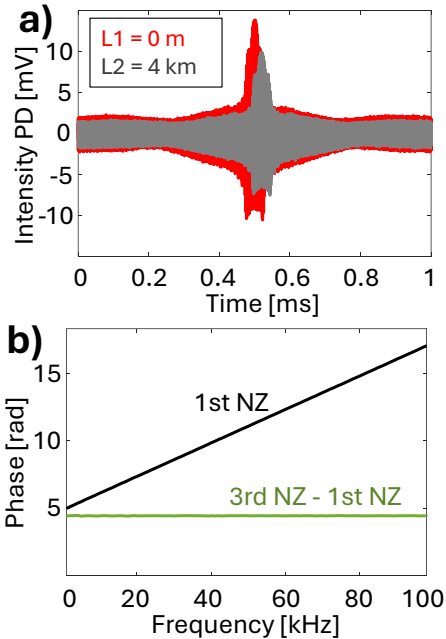


Fig. 3: Experimental results for ~ 4 km OPD. **a)** Single interferograms photodetected (PD) at both positions. The temporal shift shows a representative coarse estimation of the displacement. **b)** Phase profiles obtained from the difference of positions. In black, the raw profile from 1st NZ, being the slope the useful information. In green, the profile obtained from 3rd NZ after subtracting the one from the 1st NZ.

summarizes the OPD measurements obtained with the proposed method and compares them with independent reference measurements. For fiber lengths up to 15 m, the reference values were obtained using a LUNA OBR system with a resolution of 0.6 mm, while fibers with longer distances were characterized using an OTDR (EXFO MaxTester-730C) with spatial resolution of approximately 1.1 m, 1.2 m and 5 m for measurement ranges of 500 m, 4 km and 50 km, respectively. The good agreement between different measuring systems is shown as well in Tab. 1. It should be noted that both positive and negative OPD variations are reported, demonstrating the capability of the system to retrieve the sign of the displacement.

The performance of the dual-comb system is defined by a NAR of 102.13 km, determined by δf , while the resolution is experimentally estimated from repeated measurements under zero-displacement conditions, i.e., by acquiring multiple measurements at a fixed position and evaluating the resulting OPD, which is ideally zero. The set of pairs of measurements shows a standard deviation of 5.6 mm.

Tab. 1: Ref. OPD: Fiber OPD measured with LUNA (0 m-15 m) and OTDR (500 m, 4 km, 50 km) that serves as reference; **CPR Length:** Fiber OPD calculated with the proposed Common-Path Ranging system; **Relat. Dev.:** Relative deviation between measured and calculated lengths.

Ref. OPD	CPR Length	Relat. Dev.
0 mm	1.17 mm	1.17 mm
-17.7 cm	-17.785 cm	0.37 %
1.018 m	1.01845 m	0.033 %
-14.792 m	-14.79174 m	0.00041 %
521 m	521.3253 m	0.062 %
-4.037 km	-4.037339 km	0.0084 %
50.444 km	50.4473560 km	0.0067 %

Conclusions

The proposed ranging approach combines the intrinsic phase-noise suppression of the common-path configuration with a multi-Nyquist-zone processing strategy, enabling simultaneous access to large measurement range and high resolution.

Experimentally, optical path differences up to 50 km have been measured with a resolution of 5.6 mm, corresponding to a resolution-to-non-ambiguity-range ratio of $\sim 10^7$, using an extremely simple setup and a low-coherence optical source. By combining coarse estimation from the first Nyquist zone with fine resolution from higher-order zones, the proposed method highlights the potential of common-path architectures for scalable and long-haul fiber metrology, paving the way for implementations in distributed sensing and large-scale monitoring systems.

Acknowledgements

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